

Ranges Delay/Pulse Generator Replacement

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Jason Lamb, AF Project Manager
Mat Pilcher, BNH Project Manager
Thomas Eggenpiller, Technical Lead
E. Al Jones, System Engineer
Sloan Brown, System Engineer
Tabitha Green, System Engineer

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1. SCOPE

1.1 Background

The AEDC Hypervelocity Aeroballistic Ranges provide the ability to launch projectiles into an environmental tank for the purpose of studying various aspects of hypersonic vehicle characteristics and hypersonic impact events. Multiple types of instrumentation are required to successfully conduct testing, each with their own trigger needs. Delay generators are utilized in conjunction with Model Detection Systems to provide precision triggers to various instrumentation, including flash x-ray systems, laser photographic systems, and others. No analysis is required for delay generators as they operate as a control system.

1.2 Statement of Need

The objective of the Ranges Service Life Extension Program (SLEP) is to replace failing and obsolete systems. The goal of the Ranges Delay Generator Replacement subproject is to replace delay generators, networking hardware, and to develop a new Human-Machine Interface (HMI) to set up and control the delay generators. The legacy systems have shown an increase in component failures due to age and are far past service life. The legacy systems are no longer supported by the original equipment manufacturer (OEM).

1.3 Justification

Procuring new systems will reduce maintenance costs by having newer equipment. Procuring new systems with an increased number of outputs per generator will allow for more options and capability during routine testing. Procuring commercial-off-the-shelf (COTS) products will ensure ease of use and support from the OEM. Procuring enough systems will ensure SLEP KPPs are met, including increased throughput in Range G and Range I.

1.4 System Description

1.4.1 Description

The delay generators needed for testing are comprised of high-resolution pulse generators with fiber optic inputs and outputs. An (HMI) is required for programming and executing checkout activities. Delay times are pre-programmed by the System Engineer prior to checkouts and test events.

1.4.2 System Hierarchy

For the system hierarchy below, each asset is allocated under the process which is in the HYPERSONIC area.

Area = HYPERSONIC

Process = RANGE SUPPORT

Parent Asset = RANGE TEST CTRL

Child Asset = GI VIES

1.4.3 System Interface Triggers

The system shall provide triggers over fiber optic signals to various equipment in the G/I Screen Room and Ranges G and I test areas. System triggers will interface with the existing Ranges fiber optic network.

2. APPLICABLE DOCUMENTS

2.1 General Documents

- TOS Procedure P14-0002 – ID&C Development and Maintenance Revision 0, July 6, 2017
- NFPA 70 National Electrical Code® (NEC®) Softbound and Handbook Set, 2014 Edition

2.2 System Safety Hazard Analysis

- G RANGE VELOCITY INDENDENT EVENT SYNCHRONIZER REV. 7

2.3 Drawings

- VG532645.01 VELOCITY INDEPENDENT EVENT SYNCHRONIZER (VIES) BOM, RACK LAYOUT & INTERCONNECT VKF – G-RANGE
- VG532665.01 VIES FIBER OPTIC INTERFACE CARD PC00591 LAYOUT & SCHEMATIC VKF – G-RANGE
- VG532665.02 VIES FIBER OPTIC INTERFACE CHASSIS FAB., SILKSCREEN & ASSEMBLY VKF – G-RANGE
- VG532665.03 VIES FIBER OPTIC INTERFACE CHASSIS WIRING VKF – G-RANGE

2.4 Work Instructions

- W14-2264-3402 Time Measurement System Preventative Maintenance Activities
- W14-2264-5092 Range G Pre-Tank Closing Operations
- W14-2282-2650 Range I Pre-Tank Closing Operations
- W14-2283-2950 Range S1 Pre-Tank Closing Operations
- W14-2282-2661 Final System Check/Countdown For Range I
- W14-2264-6002 Final System Check/Countdown For Range G
- W14-2282-2661 Final System Check/Countdown For Range S1
- W14-2285-5503 VIES II Setup

3. REQUIREMENTS

3.1 System Functional and Performance Requirements

3.1.1 Specifications

The delay generators needed for testing in Ranges must meet certain requirements to effectively provide the necessary capability. The new delay generators and network switch must meet or exceed legacy specifications, including:

- Delay generators shall contain at least 1 x 1300nm fiber optic input and 8 x 820nm fiber optic outputs.
- Delay generator resolution shall be 5 nanoseconds or less.
- Delay generator shall have a LAN and GPIB port.
- Delay generator shall have channel multiplexing.
- Delay generators shall be networked to the existing VIES computer in the G/I Screen Room.
- An HMI shall be provided on the existing VIES computer.
- HMI shall allow the user to pre-program delay generator channels, pulse durations, and times remotely from the VIES computer.
- Hardware for delay generators and network switch should fit inside the legacy Range G/I Screen Room 19” network racks.
- Uncertainty analysis shall be provided
- Delay generators shall be calibrated per Air Force Technical Order at the direction of the Air Force Metrology and Calibration Authority
- An artifact of the programming shall be provided that shows what delay times were utilized on each test event

3.2 Physical Characteristics

3.2.1 Weight

The delay generators and network switch shall not exceed 10 lbs. for any single component.

3.2.2 Dimensions/Volume

The delay generators and network switch shall fit into a standard network rack, 19” width x 40U height.

3.3 Life Cycle/Duty Cycle

The life cycle is approximately 15 years. The duty cycle per system is approximately 5 times per week.

3.4 Spare Parts

A spare parts inventory of 10% spares, with a minimum quantity of one of each critical component shall be purchased. Existing central warehouse (CWH) items will be evaluated for adequate levels.

3.5 Constraints

System must be able to operate remotely.

3.6 Environmental Conditions

3.6.1 Temperature

Equipment will be used inside a building that is temperature controlled.

3.6.2 Environmental Condition, Dust, Water

Installation is in the G/I Screen Room. Dust is an environmental concern for the electronics. Power supplies and other electrical equipment must be kept off the ground if possible. Preventative Maintenance activities shall be developed to ensure the equipment remains clean and operational.

3.7 Safety

Exposure to facility 120VAC can lead to shock or electrocution.

3.8 Interoperability

The Ranges currently uses Berkeley Nucleonics delay generators exclusively. While other brands may meet specifications, attempting to integrate or mix/match other brands of delay generators can be problematic during and after integration into the testing network and HMI. This could negatively impact the test operation itself and/or the data being collected.

4. VERIFICATION/VALIDATION

All received components will be thoroughly inspected and inventoried. Resolution will be verified with the Time Measurement System (TMS). Remote triggering will be verified in conjunction with Ranges Model Detection Systems, Flash X-Rays, and Laser Photographic Systems.

Description	Method			
	A	I	D	T
20 delay generators shall be installed in the Range G/I Screen Room.		X		
Delay generators shall have a minimum of 8 fiber optic outputs.		X		
Delay generators shall be networked to the existing VIES computer.		X		
Delay generators shall have a resolution of 10ns or smaller.		X		
Delay generators shall have at least 1 fiber optic input.		X		

Delay generator fiber optic inputs and outputs shall interface with existing Ranges fiber optic network.	X		
Human-Machine Interface shall be installed on the existing VIES computer.	X		
Human-Machine Interface shall have the ability to program all delay generators remotely.		X	
Human-Machine Interface shall have the ability to trigger all delay generators remotely.		X	
Hardware for delay generators and network switch should fit inside the legacy Range G/I Screen Room 19" network racks.	X		

5. NOTES

List of Acronyms

AEDC – ARNOLD ENGINEERING DEVELOPMENT COMPLEX

CWH – CENTRAL WAREHOUSE

COTS – COMMERCIAL OFF-THE-SHELF

LGG – LIGHT-GAS GUN

SLEP – SERVICE LIFE EXTENSION PROGRAM

TMS – TIME MEASUREMENT SYSTEM

TTL – TRANSISTOR-TRANSISTOR LOGIC